

D365 F & O Warehouse Management Systems (Advance warehousing)

A Warehouse Management System – WMS – is critical to any business looking to reduce errors, keep costs in line, and increase efficiency within warehouse operations. It is all about removing complexity. A serious challenge in an area loaded with inter-connected processes, varying interfaces, and business-critical stakeholders.

Key functions in Advanced Warehouse Management:

- Mobile device – for performing work orders and scanning barcodes enforcing efficient handling when moving goods in-between locations or picking for production/sales orders.
- Work orders – System or user-directed work for managing various inbound, inventory, and outbound operations.

Core operations

- Purchase receive – Register the arrival of items on the inbound dock
- Purchase put-away – Pick registered items on the inbound dock and put them to a defined inventory location.
- Pick to production – Pick items from the inventory location and put them to the production input location.
- Production put-away – Pick items from the production output location to a defined picking location.
- Sales picking – Pick items from the picking location to the outbound dock.

Additional operations

- Cycle count – Planned or spontaneous counts of on-hand inventory on locations within the warehouse.
- Replenishment – Stock relevant picking locations based on replenishment strategies e.g., minimum and maximum stocking limits.
- Adjust inventory – Adjust inventory up or down to align system data with the physical inventory.

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- Quality orders – Perform quality tests on goods arriving at the docking area, after internal production, or before shipment.

Basic setup:

- To use Advanced Warehouse Management the following settings must be enabled.
- Activate 'Use warehouse management processes' in the described warehouse.
- Use the path: Warehouse management > Setup > Warehouse > Warehouses.

The screenshot displays the 'Warehouses' setup page in Microsoft Dynamics 365 Finance and Operations. The breadcrumb navigation at the top shows the path: Warehouse management > Setup > Warehouse > Warehouses. On the left, a list of warehouses includes 0045-Site, 0045-Trans, 007A, 007ADV (selected), 007Q, and 007Trans. The main area shows the configuration for '007ADV' (007 Advance warehouse). A red box highlights the 'Use warehouse management processes' toggle, which is set to 'Yes'. Other visible settings include 'Reserve when orders are released' (No), 'CHECK DIGITS' (No), 'REDUCE PICKED QUANTITY' (No), 'NEGATIVE INVENTORY' (No), and 'INVENTORY' (Default inventory status ID).

Warehouse	Name
007ADV	007 Advance warehouse

Warehouse
Use warehouse management processes
☒ Yes
Reserve when orders are released
☐ No
Requirement for inventory reservation
Allow partial reservation
Print BOL before confirming shipment
☐ No
Work processing policy name
Auto update shipment
On quantity decrease
Financial negative inventory
☐ No
Physical negative inventory
☐ No
CHECK DIGITS
Use unique check digits
☐ No
REDUCE PICKED QUANTITY
Decrement load line
☐ No
INVENTORY
Default inventory status ID
LABOR ST
Allow labor
☐ No
Enable quarantine
☐ No
PRODUCT
Backflush
☐ No

Note- This was a general description or an small overview about WMS and I will explain it more below.

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Sites

A site is a grouping of business resources, such as warehouses and production units, which share interrelated transactions, are located at a particular geographic location, and belong to a single legal entity. A single site cannot be shared by multiple legal entities

How to create a Site

Navigation: **Inventory management > Setup > Inventory breakdown > Sites.**

OR

Warehouse Management > Setup > Warehouse > Sites

Click on New

Finance and Operations Warehouse management > Setup > Warehouse > Sites

Save + New Delete Options

Filter

Standard view

Sites

Site Name

Test Test Abhishek

Test

Test Abhishek

General

ORDER ENTRY

Order entry deadline group

Time zone

(GMT+08:00) Beijing, Chon...

MASTER PLANNING

Use transfer journals for movement...

No

Use transfer journals for movement within Site – IF this is enabled then transfer order will not be created. It will always be a transfer Journal when the planned transfer order is confirmed.

Time zone - Select the time zone that applies to the selected site. The time zone information is used to calculate and display order entry deadlines for sales orders, sales quotations, and transfer orders in the time zone of the shipping location.

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You can add multiple address according to Business purpose

Addresses

[+ Add](#) [Advanced](#)

Name or description	Address	Purpose	Primary
			

[show here.](#)

New address

Location ID

Name or description

Purpose

[show here.](#)

Role	Description
☐ Business	Business
☐ Consignment	Consignment
☒ Delivery	Delivery
☐ Fixed asset	Fixed asset
☐ Head company	Head company
☐ Home	Home

[Select](#)

Warehouse

Default inventory status ID

Allow users on mobile devices to..

☐ No

Allow users on mobile devices to receive at another warehouse

Select the option to allow warehouse workers to receive loads and items at any warehouse in the site. Otherwise, only receipt in the warehouse specified on the source document is possible. Receiving in a different warehouse will move all processes to the new warehouse. The source document will not be updated.

After you have created a site, you can delete it, provided that the following conditions are met:

- The site is not referenced from warehouses, production resources, or other master data.
- No open on-hand inventory is associated with the site.

If either of these conditions exists, a message is displayed and you cannot delete the site.

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Warehouse

A primary function of a warehouse is offering storage space for inventory, equipment or other items. This provides a facility where businesses can store their goods when they do not need them. It also prevents stock wastage and ensures that their stock is protected and safe.

How to create a warehouse

Navigation: **Inventory management > Setup > Inventory breakdown > Warehouses**
OR
Warehouse Management > Setup > Warehouse > Warehouses

Click on New

Note – When creating a new warehouse do not press save until you have not enabled warehouse management parameter if you want to use WMS.

Finance and Operations | Warehouse management > Setup > Warehouse > Warehouses

← Edit + New Delete Warehouse Options

Standard view ▾

Warehouses

Warehouse	Name
Test	Test

General

Site	Quarantine warehouse	Goods in transit warehouse	REFERENCE
1			Vendor account
Type	Transit warehouse	Under delivery warehouse	
Default			

Type

This option allows you to specify the type of the warehouse. A warehouse of type Default can have other types of warehouses associated to handle certain warehouse processes, such as Quarantine, Transit, Goods in Transit and Under delivery.

Quarantine warehouse

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Each warehouse must be associated with a separate quarantine warehouse before you can create quarantine orders.

Transit warehouse

A transit warehouse is used for tracking in-transit transfer orders between warehouses. This helps you keep track of on-hand information when goods are in-transit. For example, orders that are in-transit between warehouses will appear on-hand in the transit warehouse.

Master planning		
ITEM COVERAGE	MAIN WAREHOUSE	Main warehouse
Manual	Refilling	
☒ No	☒ No	
Calendar		Warehouse level
		0

Manual

Select this option to manual plan item requirements. Clear this check box to automatically include items in the warehouse when planning item requirements.

Refilling

This option controls how to determine item requirements when there is no item coverage defined for the item. Select this option to explode demand based on the refill relation. Set this option to No to use the default order type for the item.(Auto planned order creation if enabled in case of Min Max)

Main Warehouse

If filled with the warehouse it will be default warehouse to refill the stock

Inventory and warehouse management			
Default receipt location	Default production input location	Default shipment maintenance lo...	Work processing policy name
Default issue location	Issue status after raw material pi...	Default location for credit only re...	Default quality maintenance loca...
	Picked		
Default production finished goo...	Default kanban finished goods lo...	Default container type	Default location for final shipment
			Default location for sorting
			Counting reason code policy

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Default receipt location

The location that will be used when on-hand inventory is registered during the inbound process. This is also the location that will be used for transactions with items that use a storage dimension group with the active location dimension. If you use the warehousing app, the location must be license plate controlled.

Default issue location

The location that is used during the outbound process. This is also the location that will be used for transactions with items that use a storage dimension group with the active location dimension. This setting will be ignored if you use advanced warehouse processes.

Default production finished goods location

When a production order is completed, this is the location that will be used for the finished goods

Default production input location

This is the location where raw material consumption will be done from.

Issue status after raw material picking

Select the status to be used when raw material picking has been completed. If you select Picked, the physical on-hand inventory will be reduced. If you select Reserved physical, the material will appear as physical on-hand inventory that is reserved at the input location where the warehouse work was completed.

Default kanban finished goods location

When a kanban order is completed, this is the location that will be used for the finished goods.

Default container type

Select the default container type to be used on license plates. Container types are used to define the physical characteristics of a tangible carrying unit, such as cage, box, pallet, or a shipping container.

Default quality maintenance location

The location specified here will be used when the quality check is rejected, and work is not created. The inventory will be immediately moved to this location and the quality order will be updated to use this location.

Default location for final shipment

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Specify the default final shipment location to be used in the container packing process when the container packing policy Warehouse selection is set to Warehouse group or All.

Default location for sorting

Specify the default sorting location to be used in the container packing process when the container packing policy Warehouse selection is set to Warehouse group or All.

Counting reason code policy

The counting reason code policy assigned to the warehouse.

Location names			
AISLE	RACK	LEVEL	POSITION
Include aisle	Include rack	Include shelf	Include bin
☐ No	☐ No	☐ No	☐ No
	Format	Format	Format
			EXAMPLE
			Location

Include aisle

Select this check box to include the name of the aisle when generating location names for new locations in the warehouse.

Include rack

Select this check box to include the name of the rack when generating location names for new locations in the warehouse.

Include shelf

Select this option to include the name of the shelf when generating location names for new locations in the warehouse.

Include bin

Select this check box to include the name of the bin when generating location names for new locations in the warehouse.

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Retail			
LOCATION REPLENISHMENT	RETURN	STORAGE DIMENSIONS	INFO CODES
Weight 0.00	Default return location 	Default location 	Reject order line
Store ☒ No			

In case of a store/outlet these parameters can also be used.

Warehouse			
Use warehouse management pro... ☒ Yes	Requirement for inventory reserv... Allow partial reservation	REDUCE PICKED QUANTITY Decrement load line ☒ No	Enable quality order for warehou... ☒ No
Print BOL before confirming ship... ☒ No	Release rule failure option for au... Warning and skip line	INVENTORY Default inventory status ID 	PRODUCTION ORDERS Backflush must use reserved qua... ☒ No
Work processing policy name 	NEGATIVE INVENTORY Financial negative inventory ☒ No	Default container type 	Posting method Inherit
Auto update shipment On quantity decrease	Physical negative inventory ☒ No	INVENTORY STATUS CHANGE Remove reservations and markings None	HAZARDOUS GOODS REPORT LANGUAGE Hazardous goods domestic repo...
RESERVATIONS Reserve only warehouse in produ... ☒ No	CYCLE COUNTING Allow license plate moves during... ☒ No	Remove inventory blocking ☒ No	Hazardous goods export report l...
Reserve inventory at load posting ☒ No	CHECK DIGITS Use unique check digits ☒ No	LABOR STANDARDS Allow labor standards ☒ No	
Reserve when orders are release... ☒ No			

Use warehouse management processes

Select the option to use the Warehousing app for optimized warehouse layout and work processing.

Print BOL before confirming shipment

Select this option if you want to print the bill of lading (BOL) before you confirm the shipment. By default, the bill of lading prints after the shipment has been confirmed.

Work processing policy name

If a policy is defined on the warehouse it will be used instead of the default policy defined on warehouse parameters.

Auto update shipment

This parameter lets you select when shipments are updated automatically. Select On quantity decrease to automatically update shipments when the quantity on the sales or transfer order

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line is decreased. Select Always to automatically update shipments when the quantity on the sales or transfer order line is decreased or increased and when new lines are added.

Reserve only warehouse in production bill of material

Warehouse management processes can be used without enabling this parameter. This option allows you to control how production bill of material items are reserved in the warehouse. Select this check box to reserve only on the dimensions above location. This enables the warehouse processes to determine location and other inventory dimensions below location in the reservation hierarchy.

Reserve inventory at load posting

Select this option to automatically reserve inventory when load is released to the warehouse.

Reserve when orders are released by a batch job

Select this option to automatically reserve inventory when a batch job is used to release orders to the warehouse.

Note- Reservation can also be controlled from other modules setups

Requirement for inventory reservation

Controls if partial release of order lines is allowed. When set to Require full reservation, order lines will not be able to be released if the total quantity is not either physically reserved or planned for cross docking

Release rule failure option for auto release

Controls how the system behaves when Requirement for inventory reservation set to Require full reservation during auto release. When Warning and skip line is selected, a warning is applied to the failed line and that line is skipped. When Error and skip order is selected, an error is applied to the order and the whole order is skipped.

Financial negative inventory

This option allows sales orders with a status of Sold to be invoiced even though financially-updated on-hand inventory does not exist. This means your inventory could go financially negative. Enabling this option is useful when inventory must be transferred between financially enabled inventory dimensions or warehouses before financially updating the inbound inventory. If this option is set to No, you cannot create an invoice even if all the inventory on-hand for the item is physically updated or received.

Allow license plate moves during cycle counting

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This option allows inventory movement to complete during counting if a license plate is in a location that is different than the location being counted.

Use unique check digits

Select this option to ensure that each location has a unique check digit.

To set up digits,

Warehouse management > Setup > Warehouse > Locations. Select Check digits.

Decrement load line

This option reduces load line quantity when the picked quantity is reduced. This is useful when a full load needs to be shipped even though the picked quantity has been reduced.

Default container type

Select the default container type to be used on license plates. Container types are used to define the physical characteristics of a tangible carrying unit, such as cage, box, pallet, or a shipping container.

Remove reservations and markings

Select what should happen to reservations and markings when inventory status changes.

Remove inventory blocking

Select this option to remove inventory blocking when inventory status changes. This option is used in quality inspections when items are automatically blocked from consumption or use and when inventory status changes after quality inspection finishes.

Allow labor standards

Select this option to allow labor standard calculations to be run in this warehouse. Setting this option to Yes can have a performance impact on the warehouse.

Enable quality order for warehouse processes

When enabled, quality orders created in this warehouse will use the quality order for warehouse processes functionality, including quality item sampling work, and quality order work. When disabled, quality order functionality will still be supported, but will not integrate with warehouse work.

Backflush must use reserved quantities

Set this option to "Yes" to only allow backflushing of reserved quantities.

osting method

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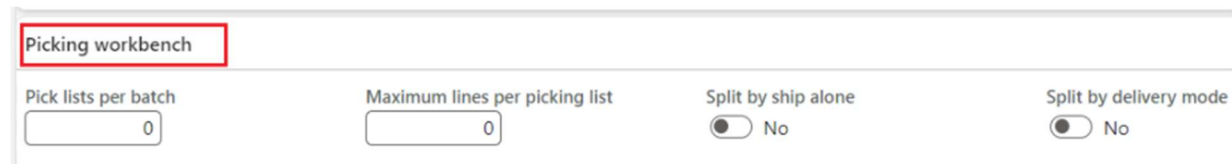
Choose whether goods reported as finished should be registered as physically available before or after the related journal postings are fully processed. Choose "Immediate" to set the system to finish all postings before it registers a finished good as physically available. Choose "Deferred" to defer the journal postings, which will ensure that physical availability of finished goods won't be impacted by these and other related processes. Choose "Inherit" to inherit this setting from the parent site or company.

Hazardous goods domestic report language

Select the default language for hazardous materials reports that are associated with domestic shipments.

Hazardous goods export report language

Select the default language for hazardous materials reports that are associated with international shipments.



The screenshot shows the 'Picking workbench' settings. The title 'Picking workbench' is highlighted with a red box. Below the title, there are four settings: 'Pick lists per batch' with a text input field containing '0'; 'Maximum lines per picking list' with a text input field containing '0'; 'Split by ship alone' with a toggle switch set to 'No'; and 'Split by delivery mode' with a toggle switch set to 'No'.

Pick lists per batch

Enter the maximum number of picking lists that can be included in a batch. If this field is left blank or zero, there will be no limit on how many picking lists can be contained in a pick wave.

Maximum lines per picking list

Enter the maximum number of picking lines that can be included in a picking list. If this field is left blank or zero, there will be no limit on how many picking lines can be contained in a picking list.

Split by ship alone

Select this check box if you want to split the picking batches for items that have been marked as ship alone on the Released products form. An item that is already packaged for shipping, such as a television, will be put on a separate picking list.

Split by delivery mode

Select this option to split the picking list for items that are in a different delivery mode. Split by delivery mode will group items together that have the same mode of delivery, such as by sea or by air.

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Note – Once the Warehouse management is enabled then other setups need to be done to process different types of orders using Warehouse management.

All setups will be warehouse and module(Process) specific.

Default setups can be initially created by system as well using below functionality.

Warehouse management > Setup > Wizards > Warehouse management initiation wizard

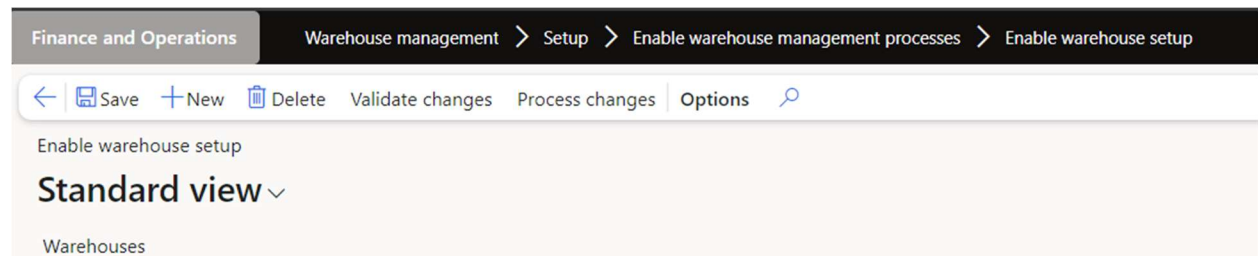
When setting up Warehouse management module, there must be multiple configurations completed. This wizard enables you to set up the following:

- Location formats
- Location types
- Location profiles
- Inventory status
- Default work user
- Mobile device menu
- Regenerate load posting methods
- Regenerate wave processing methods
- Warehouse management parameters

But this need to be updated as per the requirement of Business.(Will discussed in deep in coming Articles)

Case – If we need to enable warehouse management for already created warehouse having transactions.

Solution



Storage Dimension Group

For items to be used with the functionality of WMS they should have a dimension group which is WMS enabled.

I recommend the process of start using a Storage dimension group with the "Use warehouse

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management processes" enabled and then later use the warehouse migration process to enable the warehouses.

It is more difficult to convert items at a later date (But can be done)

Warehouse management should be switched on at the Storage Dimension Group at least, even if the customer is not yet working with the WMS.

Product information management > Setup > Dimension and variant groups > Storage dimension groups

Delete Related items Options

Standard view

Storage dimension groups

Name	Description
Ware	Warehouse Management Pro...

Storage dimensions

Name	Active	Blank recei...	Blank issue...	Physical in...	Financial i...	Coverage ...	For purcha...	For sales p...
Site	✓			✓	✓	✓	✓	✓
Warehou...	✓			✓		✓	✓	✓
Location	✓			✓				
Inventory...	✓			✓				
License p...	✓	✓	✓	✓				

Warehouse specific setup

Use warehouse management pro...

☒ Yes

WAREHOUSE SPECIFIC SETUP

Mandatory

☐ No

Primary stocking

☒ Yes

The above setup is Business dependent.

In coming Article will be sharing information about other setups and how they work for different modules.